

Using time series analysis for sales and demand forecasting

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1. Introduction

1.1 Business Context

Nielsen BookScan provides book sales data that helps publishers track performance and trends. To support independent publishers, Nielsen aims to develop a forecasting service that improves demand prediction, stock management, and investment decisions, while helping identify titles with strong long-term sales potential.

1.2 Objective of the Analysis

This analysis examined sales data for *The Alchemist* and *The Very Hungry Caterpillar* to identify demand patterns and evaluate SARIMA, XGBoost, LSTM, and hybrid models for 32-week weekly and 8-month monthly forecasting.

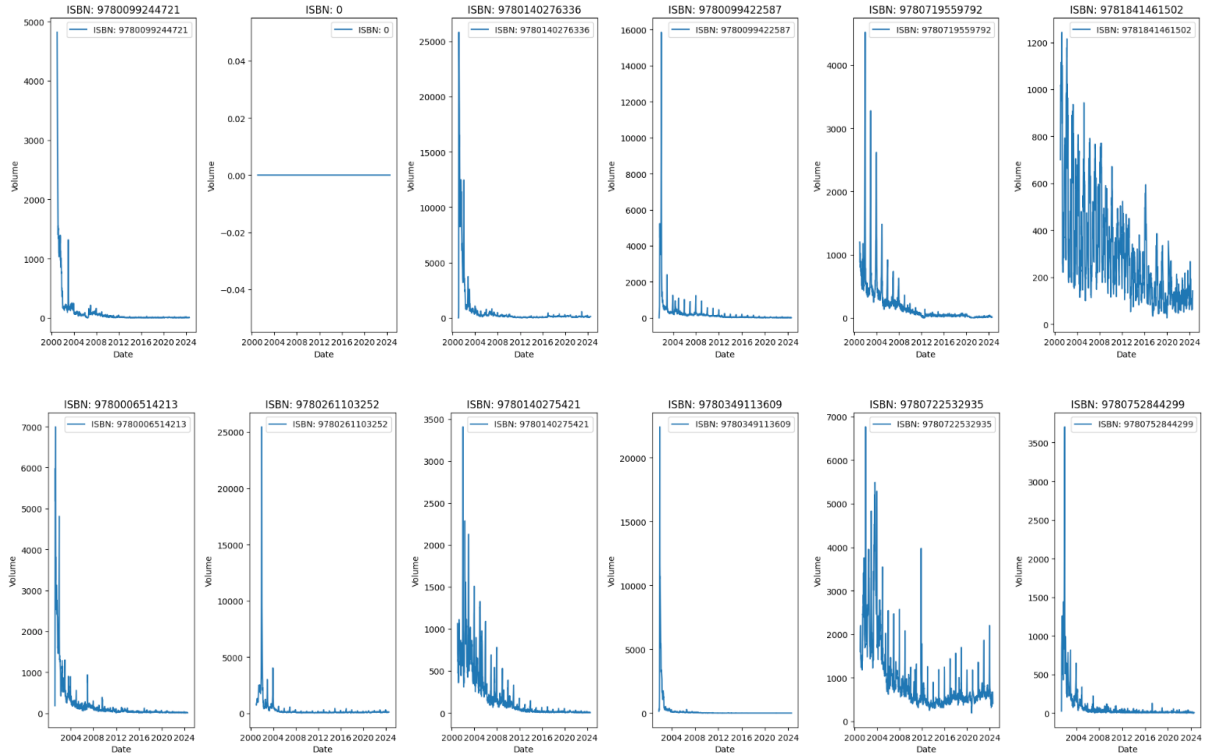
2. Data Preparation and Explanatory Analysis

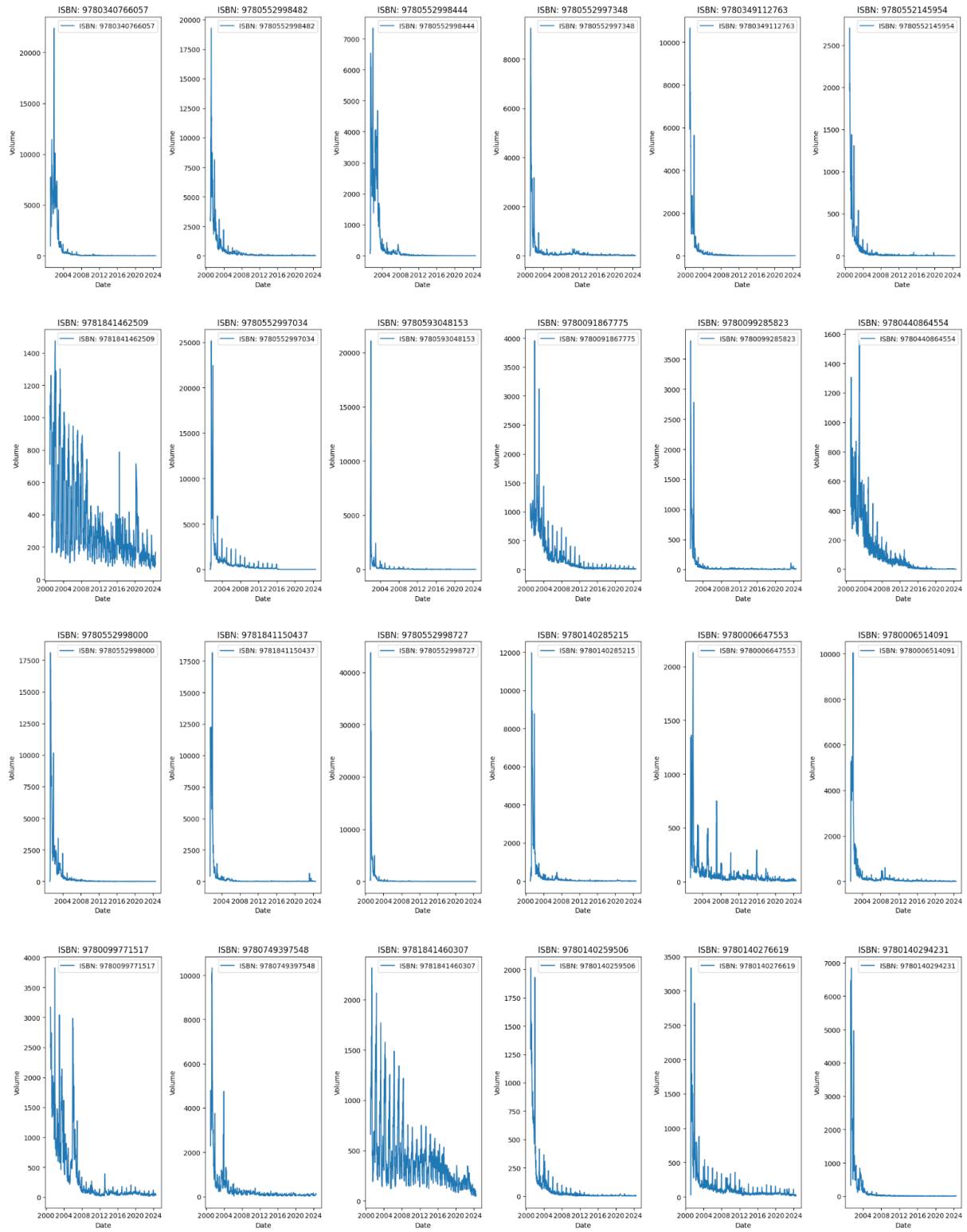
2.1 Data Preparation

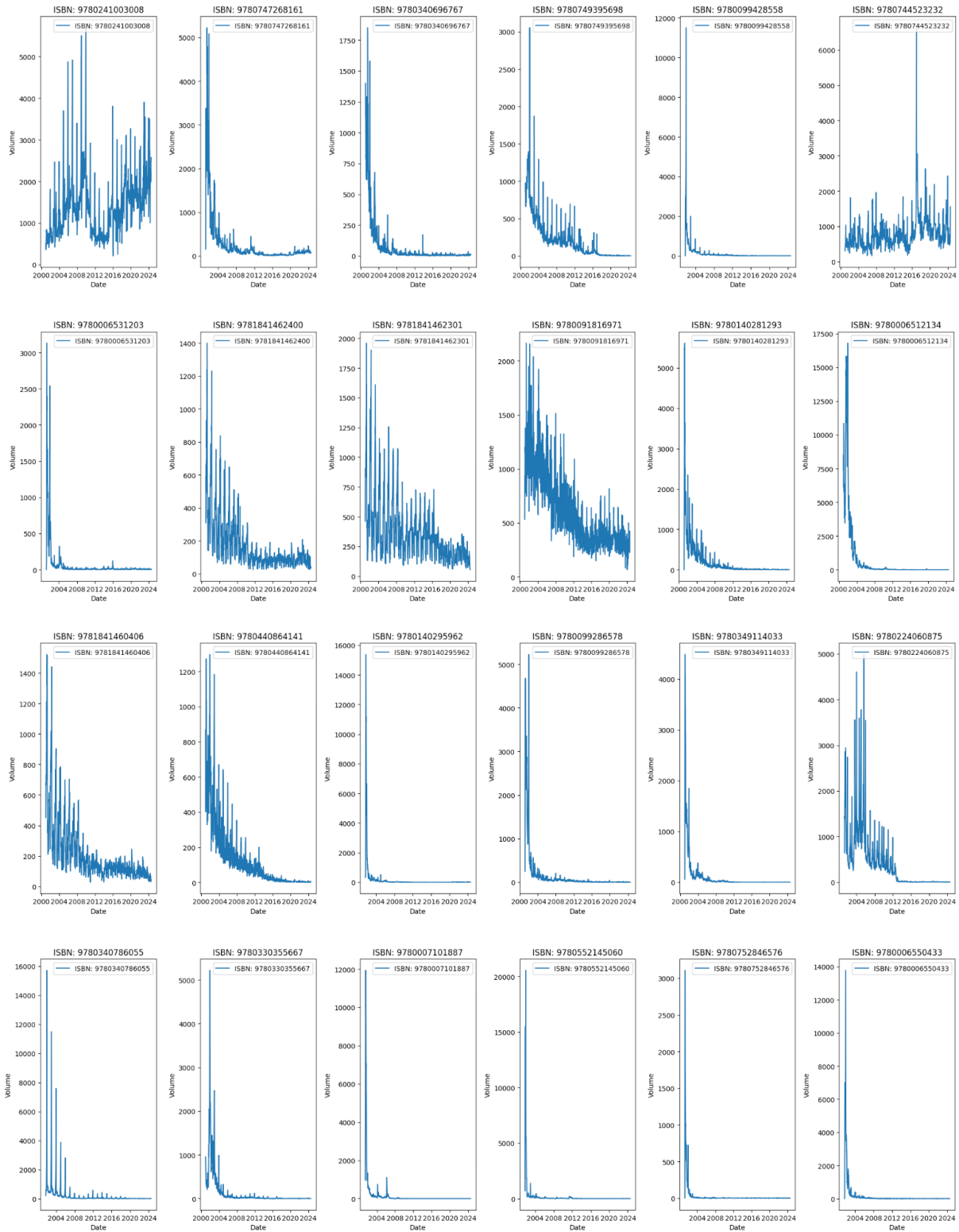
Data from both Nielsen BookScan datasets were combined and prepared for analysis by converting ISBNs and dates into appropriate formats. The data were then resampled to a weekly frequency, with missing weeks assigned zero sales to create continuous time series for forecasting.

2.2 Initial Data Investigation

Following preprocessing, 62 ISBNs with sales records extending beyond 1 July 2024 were identified and retained for visualisation and further analysis.







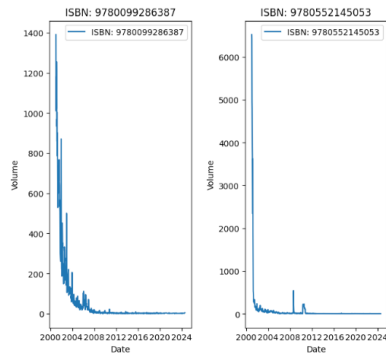


Figure 1. 62 ISBNs sales volume over time

Initial visual inspection revealed substantial variation in sales performance, with few books showing short-lived demand spikes and others exhibiting stable long-term sales and seasonal peaks.

2.3 Sales Pattern Analysis

Comparing the first 1–12 years with the subsequent 12–24 years showed that sales were generally highest early in a book’s lifecycle before gradually declining. However, few titles continued to generate long-term sales, while seasonal fluctuations persisted for several books.

2.4 Selections of Titles Forecasting

The Alchemist and The Very Hungry Caterpillar were selected for analysis, representing two distinct demand profiles within the dataset.

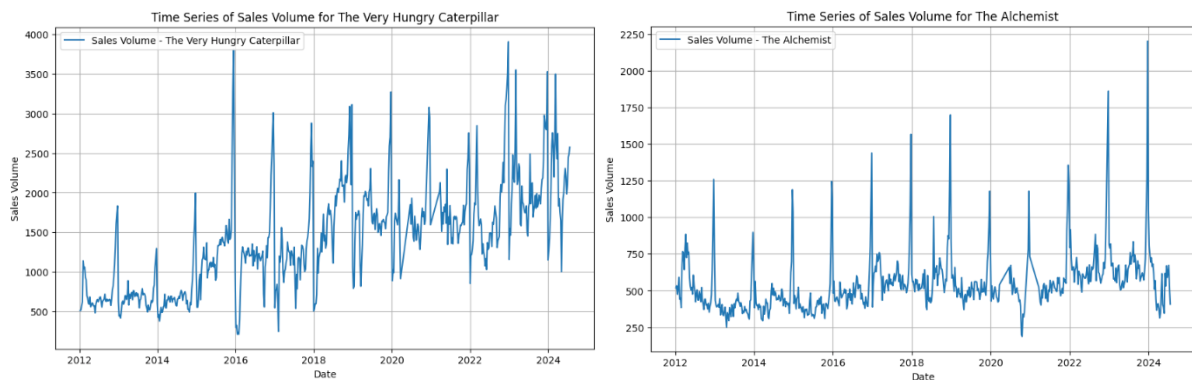


Figure 2. The Very Hungry Caterpillar and The Alchemist Weekly sales volume

The Very Hungry Caterpillar showed higher sales, greater variability, and stronger long-term growth, while The Alchemist exhibited lower sales with more consistent recurring peaks.

3. Time Series Characteristics

3.1 Decomposition Analysis

Time series decomposition of The Very Hungry Caterpillar revealed a strong upward trend, clear annual seasonality, and relatively small residual fluctuations, indicating that most sales variation was explained by trend and seasonal patterns.

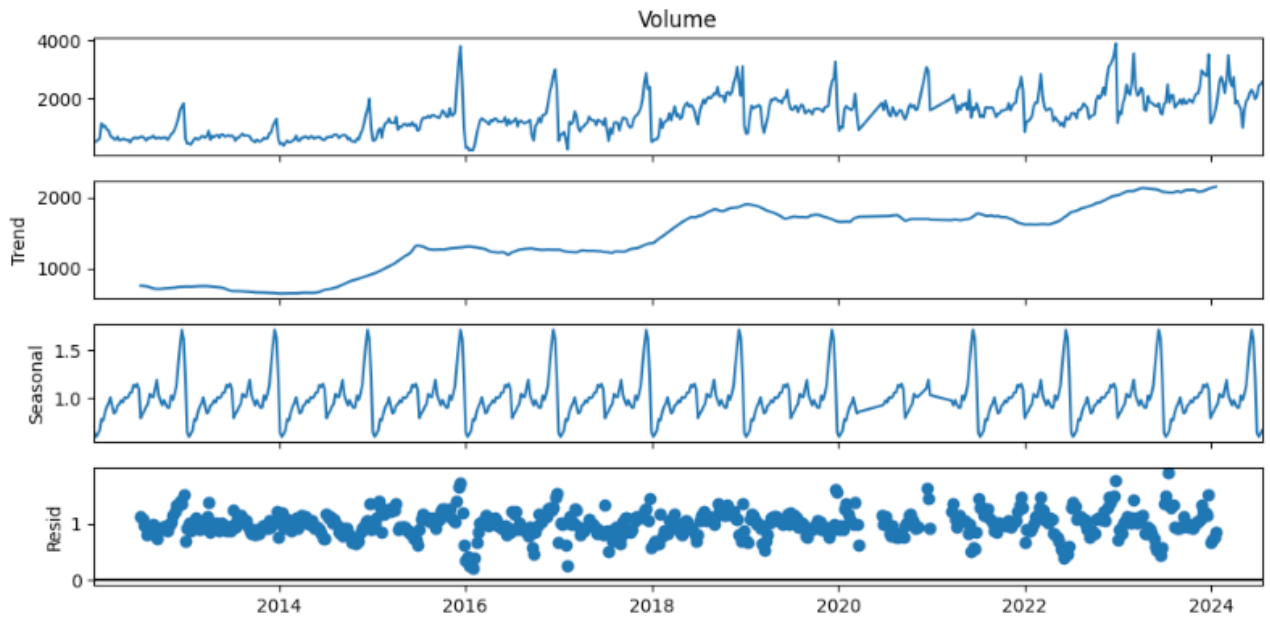


Figure 3. Multiplicative decomposition of *The Very Hungry Caterpillar*

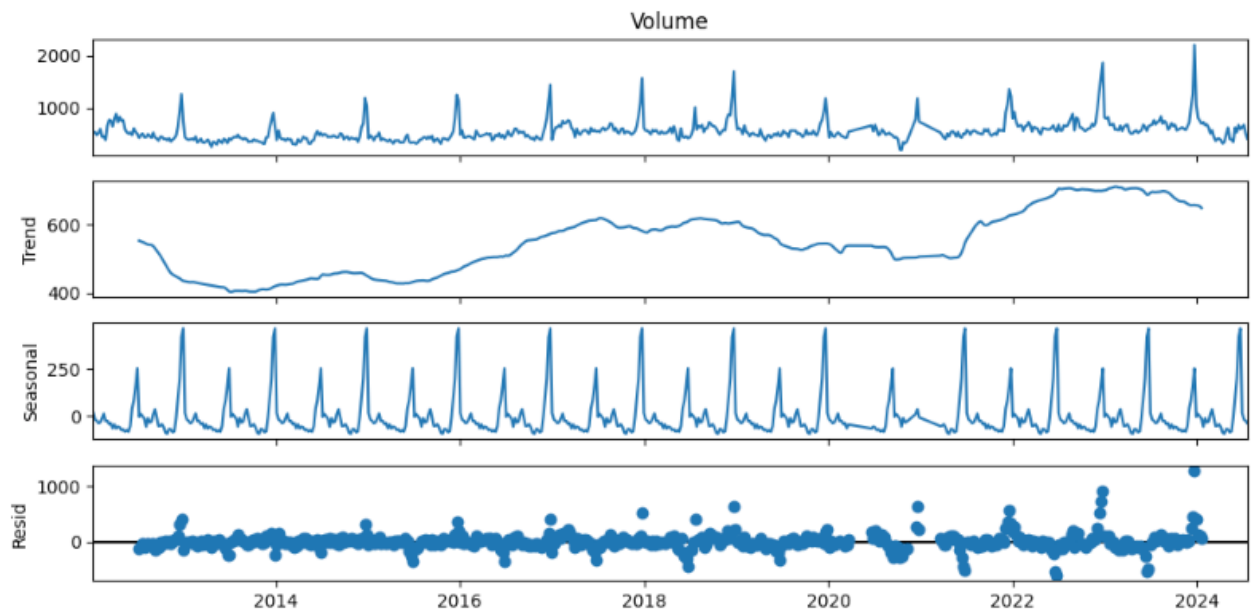


Figure 4. Additive decomposition of *The Alchemist*

For *The Alchemist*, an additive decomposition revealed a relatively stable trend with recurring seasonal peaks. Several residual outliers remained, suggesting occasional demand shocks not explained by trend or seasonality.

3.2 Autocorrelation Analysis (ACF and PACF)

The ACF plots showed strong autocorrelation across multiple lags for both books, indicating that past sales strongly influenced future demand.

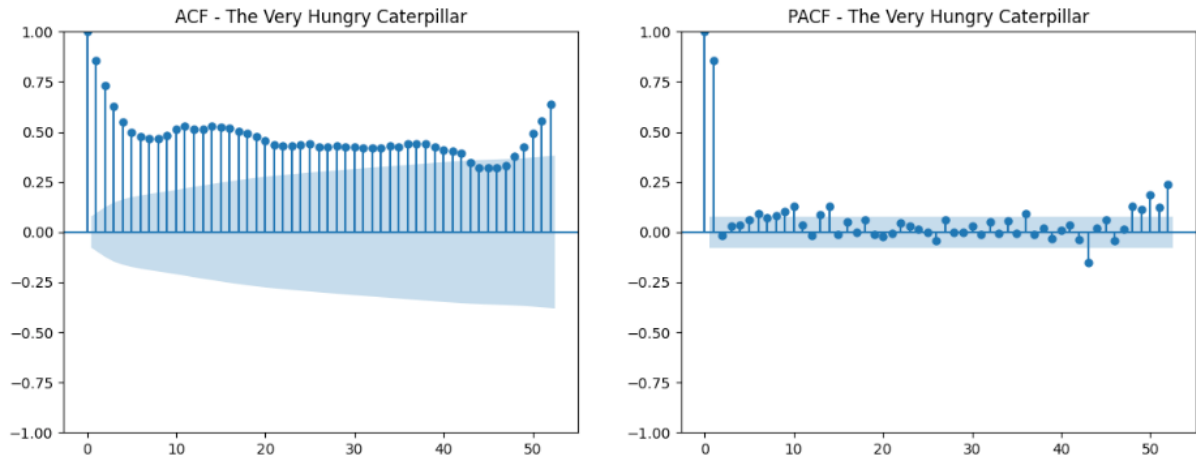


Figure 5. ACF and PACF for *The Very Hungry Caterpillar*

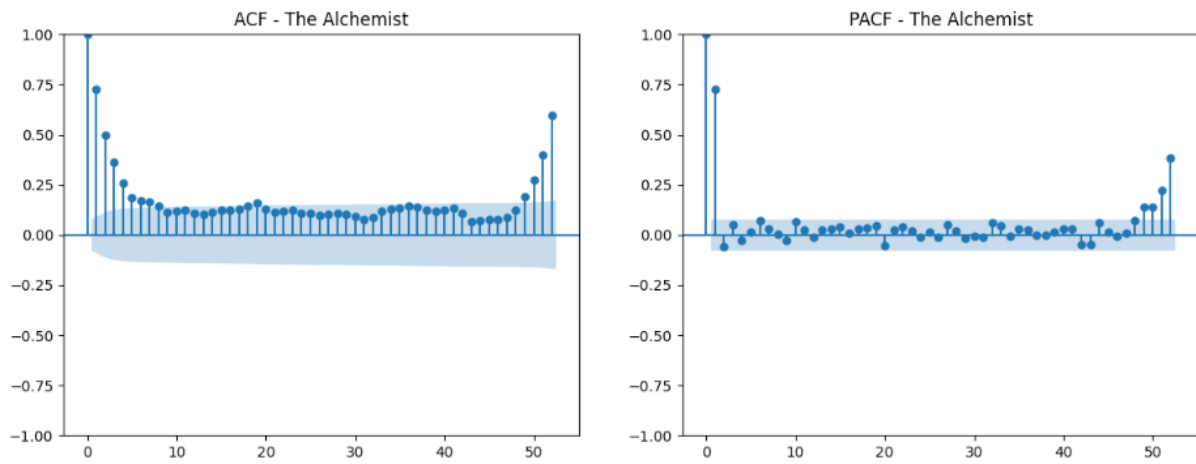


Figure 6. ACF and PACF for *The Alchemist*

The PACF plots showed a strong lag-1 effect and seasonal spikes for both books, indicating autoregressive behaviour and reinforcing the suitability of SARIMA for forecasting.

3.3 Stationarity Assessment

The ADF test showed that *The Very Hungry Caterpillar* was non-stationary ($p = 0.206$), reflecting its strong trend and seasonality, while *The Alchemist* was stationary ($p < 0.001$). These results highlight the distinct statistical characteristics of the two series.

4. Forecasting Model Development and Results

4.1 SARIMA (Auto ARIMA) Forecasting

For the 32-week forecast horizon, Auto ARIMA selected $ARIMA(1,1,1)(1,0,1)[52]$ for *The Very Hungry Caterpillar* and $ARIMA(0,1,2)(1,0,1)[52]$ for *The Alchemist*, with seasonal terms capturing annual sales patterns and differencing improving model fit.

	Book	Model	MAE	MAPE
0	The Very Hungry Caterpillar	Auto ARIMA (SARIMA)	351.196614	0.184145
1	The Alchemist	Auto ARIMA (SARIMA)	135.691503	0.229305

Table 1. SARIMA (Auto-ARIMA) MAE and MAPE

The SARIMA models delivered strong forecasting performance for both books, achieving MAPEs of 0.229 for The Alchemist and 0.184 for The Very Hungry Caterpillar. These results demonstrate SARIMA's effectiveness in capturing trend and seasonal patterns.

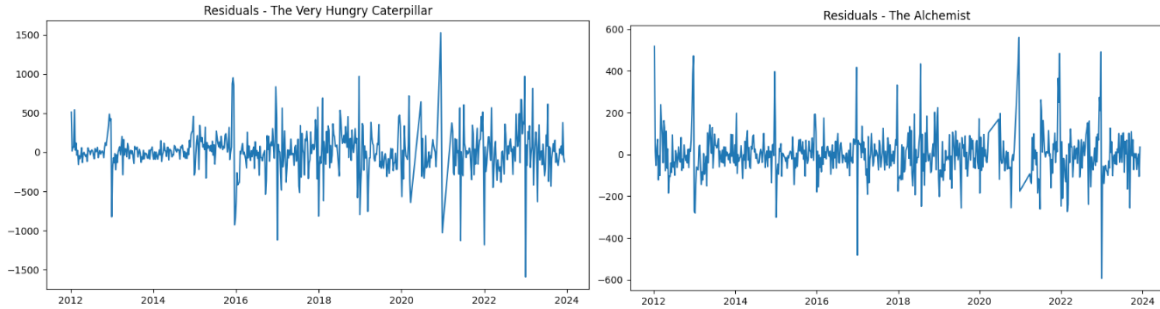


Figure 7. SARIMA residuals for both books

The residual plots fluctuated around zero, suggesting that the SARIMA models captured most of the trend and seasonality, although some unexplained volatility remained, particularly for The Very Hungry Caterpillar.

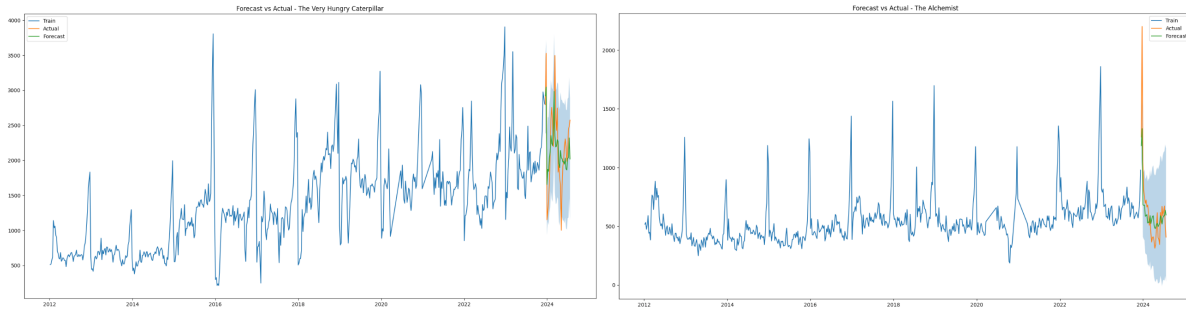


Figure 8. SARIMA Forecast vs Actual for both books

For both books, the SARIMA forecasts captured overall trends and seasonality, with most observations falling within the confidence intervals. However, the models were less effective at predicting sudden demand spikes, particularly for The Alchemist.

4.2 XGBoost Forecasting

An XGBoost forecasting pipeline was optimised using GridSearchCV. The best models used a 12-week window for The Very Hungry Caterpillar and a 6-week window for The Alchemist, with forecasts generated from lagged sales data.

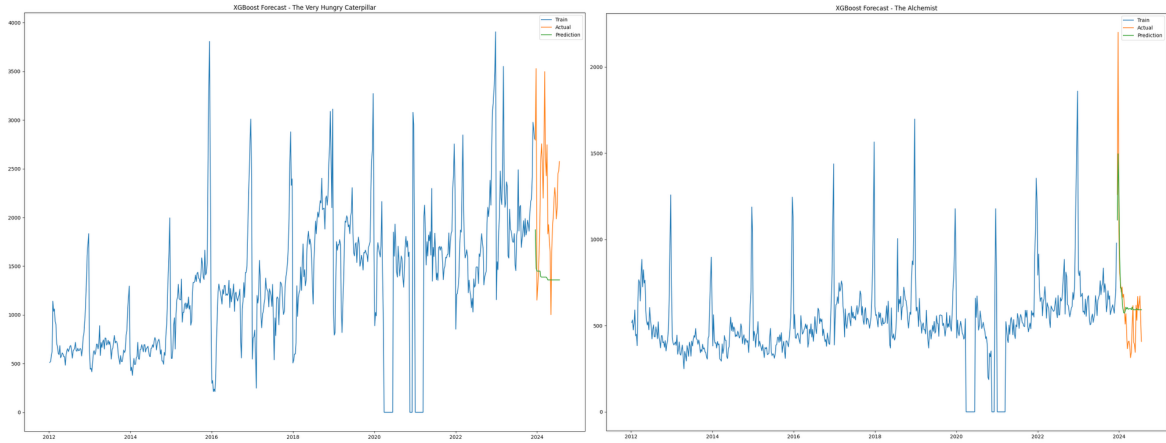


Figure 9. XGBoost Forecast vs Actual for both books

		MAE	MAPE
Book	Model		
The Alchemist	Auto ARIMA (SARIMA)	135.691503	0.229305
	XGBoost	139.315985	0.269836
The Very Hungry Caterpillar	Auto ARIMA (SARIMA)	351.196614	0.184145
	XGBoost	839.380333	0.353983

Table 2. XGBoost MAE and MAPE

XGBoost underperformed the other models, achieving MAPEs of 0.270 for The Alchemist and 0.354 for The Very Hungry Caterpillar, suggesting it struggled to capture strong seasonal patterns.

4.3 LSTM Forecasting

LSTM networks were developed to capture complex non-linear sales patterns, with KerasTuner used to optimise the model architecture and hyperparameters for each book.

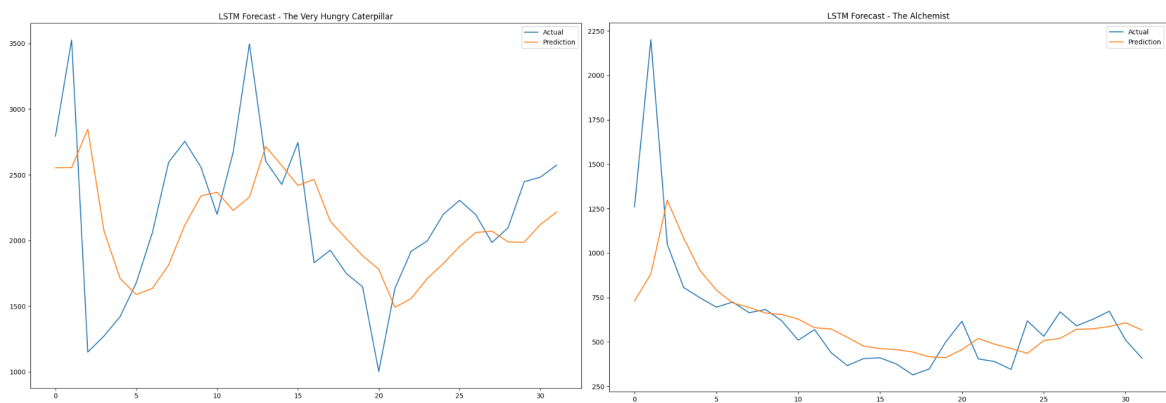


Figure 10. LSTM Forecast vs Actual for both books

		MAE	MAPE
Book	Model		
The Alchemist	Auto ARIMA (SARIMA)	135.691503	0.229305
	LSTM	152.618899	0.215461
	XGBoost	139.315985	0.269836
The Very Hungry Caterpillar	Auto ARIMA (SARIMA)	351.196614	0.184145
	LSTM	427.356236	0.226528
	XGBoost	839.380333	0.353983

Table 3. LSTM MAE and MAPE

The LSTM model produced mixed results, outperforming SARIMA for The Alchemist (MAPE = 0.215) but underperforming for The Very Hungry Caterpillar (MAPE = 0.227), suggesting that its effectiveness depended on the characteristics of the sales series.

4.4 Hybrid Model Forecasting

Sequential and parallel hybrid SARIMA-LSTM models were developed, with the sequential model achieving MAPEs of 0.236 for The Alchemist and 0.186 for The Very Hungry Caterpillar, offering only marginal improvement over SARIMA.



Figure 11. SARIMA + LSTM (Hybrid Sequential) Forecast vs Actual for both books

		MAE	MAPE
Book	Model		
The Alchemist	Auto ARIMA (SARIMA)	135.691503	0.229305
	LSTM	152.618899	0.215461
	SARIMA + LSTM (Sequential)	139.528636	0.235538
	XGBoost	139.315985	0.269836
The Very Hungry Caterpillar	Auto ARIMA (SARIMA)	351.196614	0.184145
	LSTM	427.356236	0.226528
	SARIMA + LSTM (Sequential)	354.343296	0.185938
	XGBoost	839.380333	0.353983

Table 4. SARIMA + LSTM (Hybrid Sequential) MAE and MAPE

The parallel hybrid model combined SARIMA and LSTM forecasts using equal weights, achieving the best overall performance for The Alchemist (MAPE = 0.197) and competitive results for The Very Hungry Caterpillar (MAPE = 0.190).



Figure 12. SARIMA + LSTM (Hybrid Parallel 0.5/0.5) Forecast vs Actual for both books

Book	Model	MAE	MAPE
The Alchemist	Auto ARIMA (SARIMA)	135.691503	0.229305
	LSTM	152.618899	0.215461
	SARIMA + LSTM (Parallel)	124.000562	0.197235
	SARIMA + LSTM (Sequential)	139.528636	0.235538
	XGBoost	139.315985	0.269836
The Very Hungry Caterpillar	Auto ARIMA (SARIMA)	351.196614	0.184145
	LSTM	427.356236	0.226528
	SARIMA + LSTM (Parallel)	354.086264	0.189530
	SARIMA + LSTM (Sequential)	354.343296	0.185938
	XGBoost	839.380333	0.353983

Table 5. SARIMA + LSTM (Hybrid Parallel 0.5/0.5) MAE and MAPE

Optimal SARIMA-LSTM weightings differed by book, 90% SARIMA / 10% LSTM for The Very Hungry Caterpillar and 40% SARIMA / 60% LSTM for The Alchemist.

SARIMA Weight	0.900000
LSTM Weight	0.100000
MAE	346.027521
MAPE	0.182551

Table 6. SARIMA + LSTM (Hybrid Parallel Optimised Weight) MAE and MAPE – The Very Hungry Caterpillar

SARIMA Weight	0.400000
LSTM Weight	0.600000
MAE	125.338709
MAPE	0.195793

Table 7. SARIMA + LSTM (Hybrid Parallel Optimised Weight) MAE and MAPE – The Alchemist

4.5 Model Performance Comparison

The optimised parallel hybrid model achieved the best forecasting performance for both books, with MAPEs of 0.183 for The Very Hungry Caterpillar and 0.196 for The Alchemist. This suggests that combining SARIMA and LSTM can effectively capture both seasonal and complex demand patterns.

Book	Model	MAE	MAPE
The Alchemist	Auto ARIMA (SARIMA)	135.6915	0.2239
	LSTM	152.6189	0.2155
	SARIMA + LSTM (Parallel) - 0.5/0.5	124.0006	0.1972
	SARIMA + LSTM (Sequential)	139.5286	0.2355
	XGBoost	139.3160	0.2698
	SARIMA + LSTM (Parallel) - Optimised Weight	125.3387	0.1958
The Very Hungry Caterpillar	Auto ARIMA (SARIMA)	351.1966	0.1841
	LSTM	427.3562	0.2265
	SARIMA + LSTM (Parallel) - 0.5/0.5	354.0863	0.1895
	SARIMA + LSTM (Sequential)	354.3433	0.1859
	XGBoost	839.3803	0.3540
	SARIMA + LSTM (Parallel) - Optimised Weight	346.0275	0.1826

Table 8. Models Performance Comparison

5. Monthly versus Weekly Forecasting

5.1 Monthly Forecasting Results

To evaluate the impact of data aggregation, weekly sales were converted to monthly totals and used to generate 8-month forecasts, with SARIMA and XGBoost assessed using MAE and MAPE.

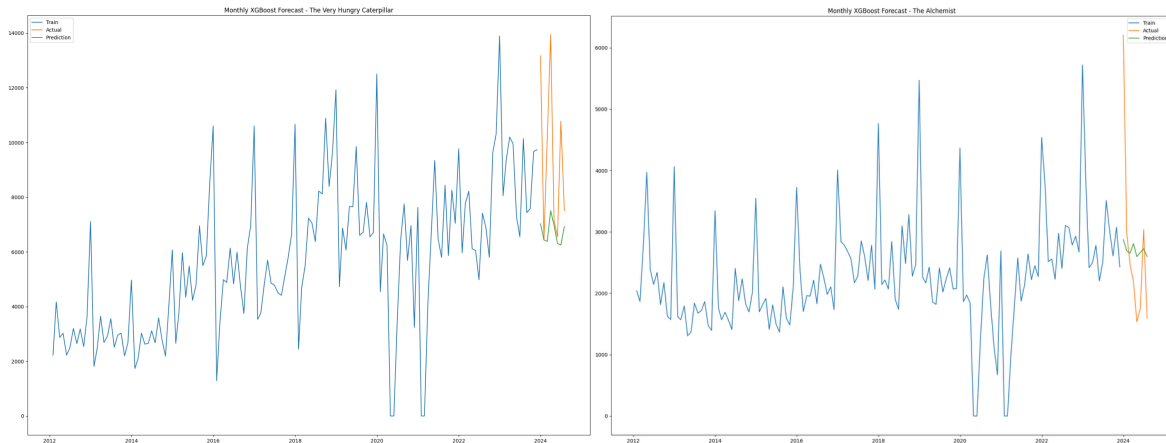


Figure 13. Monthly XGBoost Forecast vs Actual for both books

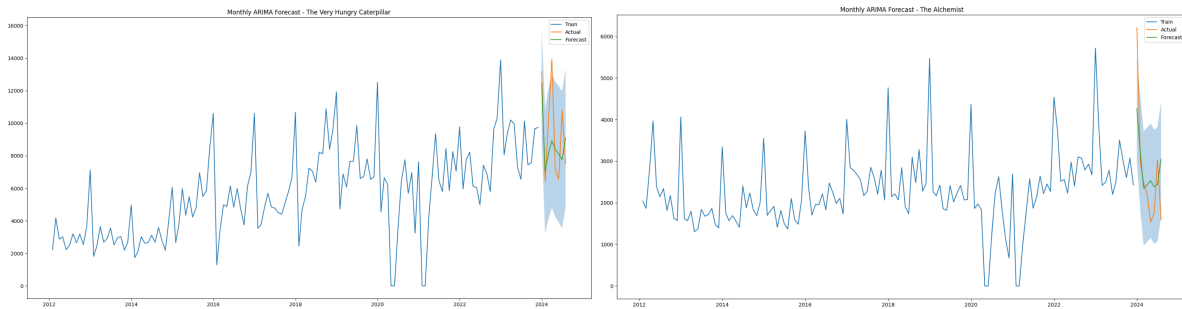


Figure 14. Monthly SARIMA Forecast vs Actual for both books

Book	Model	MAE	MAPE
The Very Hungry Caterpillar	XGBoost	2731.189758	0.232143
The Alchemist	XGBoost	962.809174	0.366757
The Very Hungry Caterpillar	SARIMA	1969.358099	0.201044
The Alchemist	SARIMA	773.087053	0.332865

Table 9. Monthly forecasting models performance

For both books, SARIMA consistently outperformed XGBoost for monthly forecasting, indicating that seasonal patterns remained a key driver of demand even after sales were aggregated to monthly totals.

5.2 Comparison of Monthly and Weekly Forecast

Comparing weekly and monthly forecasts showed that weekly SARIMA models were more accurate for both books, suggesting that weekly data better captured sales dynamics and seasonal patterns. XGBoost underperformed SARIMA at both frequencies, indicating that weekly forecasting is

more suitable for operational planning, while monthly forecasting may be better suited to strategic decision-making.

6. Key Findings and Business Recommendations

The analysis showed that *The Alchemist* and *The Very Hungry Caterpillar* exhibit sustained demand and clear seasonal patterns, making them suitable for forecasting. SARIMA delivered strong performance, while hybrid SARIMA-LSTM models achieved the best overall results. Publishers should use forecasting models to support inventory planning and reprinting decisions, increasing stock ahead of seasonal demand peaks and reducing the risk of overstocking or stock shortages.

7. Conclusion

This project demonstrated the value of time series analysis for forecasting book sales. Among the models evaluated, SARIMA and hybrid SARIMA-LSTM models delivered the highest accuracy, while weekly forecasts outperformed monthly forecasts. Overall, the findings show that forecasting can support more effective inventory planning and reprinting decisions for long-selling titles.